

Ashraf Khan

Machine Learning Engineer · Generative AI Systems Engineer · LLM Fine-Tuning & Inference Optimization

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🎯 Professional Executive Summary

Production-grade **Machine Learning & Generative AI Systems Engineer** with **5+ years of software engineering experience** and **3 published GenAI research-grade portfolio systems** with empirically verified benchmark receipts. Specializes in LLM inference optimization (4-bit NF4 quantization, LoRA/PEFT fine-tuning), agentic reasoning pipelines (multi-stage DAGs, hybrid RAG), adversarial safety guardrails, and FastAPI streaming deployments on GPU hardware (T4/L4). All systems are backed by reproducible evaluation harnesses, CI/CD automation, and measurable cost, latency, and financial impact metrics.

🤖 Generative AI & Machine Learning Portfolio Projects

1. Customer Support Automation — Quantized LLM + Agentic Workflow

GitHub: [Customer-Support-Automation-Quantized-LLM-Agentic-Workflow](#) | **HuggingFace Demo:** [Live Space](#) | **Colab:** [Interactive Notebook](#)

- **Problem:** High inference costs and latency SLAs when deploying 7B+ LLMs for high-volume tier-1 customer support, coupled with adversarial prompt injection and billing/legal safety risks.
- **4-Bit NF4 Quantization** (`bitsandbytes`): VRAM reduced from **14.54 GB → 5.09 GB (65% reduction)**, enabling low-cost T4/L4 GPU deployment.
- **Multi-Tier Safety Guardrails:** Real-time threat catalog blocking DAN-mode jailbreaks, system prompt leaks, and legal/fraud keywords (`chargeback` , `subpoena` , `GDPR`).
- **Enterprise Hybrid RAG:** BM25 keyword search fused with Dense Vector Cosine Similarity via Reciprocal Rank Fusion (RRF) for grounded, hallucination-resistant responses.
- **FastAPI SSE Streaming:** Server-Sent Events token-streaming endpoint for ultra-low Time-To-First-Token (TTFT).
- **Empirical Receipts (Verified `benchmark_metrics.json`):**
- Cost per 1,000 tickets: *0.8921* → **0.0297 (96.67% cost reduction)**

- Mean latency: **7.68x improvement** across p50/p90/p95/p99 percentiles
 - Safety Guardrail F1: **0.9412**
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2. Contract Risk Review — LoRA Fine-Tuning & Multi-Agent Legal Reasoning

GitHub: [LLM-Fine-Tuning-Agentic-Reasoning-for-Contract-Risk-Review](#)

- **Problem:** Generic LLMs hallucinate and miss liability, indemnification, and governing law risks in enterprise legal contracts — high-stakes failures with direct financial consequences.
 - **LoRA / PEFT Fine-Tuning** on DeBERTa-v3 and Qwen2.5, trained on the LEDGAR legal corpus for deterministic clause risk classification.
 - **ChromaDB Precedent Retrieval:** Vector embedding retrieval grounding risk determinations against statutory definitions and contract playbooks.
 - **Fairness & Bias Auditing:** Rigorous evaluation across contract jurisdictions and party asymmetries.
 - **Empirical Receipts:**
 - Clause Risk Classification F1: **1.000** (balanced LEDGAR evaluation set)
 - Precision liability, indemnification, and governing law extraction with near-zero hallucination rate on clause boundaries
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3. Quantum Hardware Validation & Benchmarking Suite

GitHub: [Quantum-Hardware-Validation-Benchmarking-Suite](#)

- **Problem:** No standardized empirical reporting framework for evaluating NISQ circuits and algorithmic fidelity across realistic hardware error models.
 - **Hardware-Agnostic Benchmarking Engine:** Profiles quantum circuit depth, gate counts, and decoherence times (T1, T2).
 - **Noise Simulation & Error Mitigation:** Evaluates Zero-Noise Extrapolation (ZNE) and readout error mitigation strategies.
 - **Automated Empirical Reporting:** Publishes fidelity decay curves, error benchmarking results, and algorithmic profiling dashboards.
 - 100% passing Pytest unit & integration suites with reproducible `benchmark_metrics.json` receipts.
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Professional Experience

Senior Software & Systems Engineer | Logicloop (Mumbai, India | Jan 2026 – Present)

- Built MAXX AI Voice Assistant with custom wake-word activation and multi-advisor conversational flows via Millis.ai (LLM integration).
- Led 3 developers; enforced CI/CD automation and Level 2 VAPT security standards across enterprise financial flows.

Full-Stack Developer & Data Custodian | Fitwell Technologies Inc. (Remote – VA, USA / Jun 2025 – Present)

- Integrated LLM-powered features, Amazon Connect voice AI workflows, and HIPAA-compliant data pipelines on AWS private cloud.
- Deployed self-hosted Supabase PostgreSQL cloud backend on EC2 for PHI data isolation; maintained zero data breach posture.

Lead Software Engineer | QuantGen Private Limited (India / Nov 2024 – Jun 2025)

- Delivered AI-integrated health platform (FitWell) and enterprise 24/7 operations platform (Kokomo) with AWS ECS/App Runner orchestration.

Software Engineer | Logicloop (Mumbai, India / May 2024 – Oct 2024)

- Built Flutter frontend for HDFC Life InstaQuote, connecting to insurance pricing ML backend APIs.

Software Engineer | SoftProdigy (Remote – USA / Nov 2023 – Apr 2024)

- Developed loan lending application workflows and implemented gRPC client API libraries.

Senior Software Development Engineer | AKcess Labs Pvt. Ltd. (Remote – London, UK / Jan 2023 – Nov 2023)

- Built Flutter cross-platform applications with Flask backends for hyperlocal commerce platform.

Software Developer | We3.Tech (Maharashtra, India / Aug 2021 – Jan 2023)

- Delivered eKYC, Digio eSign integrations, and banking API modules for Motilal Oswal MO Investor (SEBI/RBI compliant cryptographic security).

Technical Skills & Core Competencies

- **LLM Optimization & Inference:** 4-bit NF4 Quantization, `bitsandbytes`, GGUF, Torch `float16`, VRAM profiling, T4/L4 GPU deployment.
- **Fine-Tuning & PEFT:** LoRA, QLoRA, PEFT Adapters, DeBERTa-v3, Qwen2.5, LEDGAR Corpus, Hugging Face `transformers`, `trl`.
- **Agentic Pipelines & RAG:** Multi-stage DAGs, ChromaDB, BM25 + Dense RRF Hybrid Search, Reciprocal Rank Fusion, LangChain, Prompt Engineering.

- **Safety & Adversarial Defense:** Multi-tier guardrail catalogs, DAN/jailbreak interception, GDPR/legal keyword routing, human escalation DAGs.
 - **APIs & Serving:** FastAPI (Async), SSE token streaming, REST, gRPC, Protobuf, WebSockets.
 - **ML Frameworks:** PyTorch, TensorFlow, scikit-learn, OpenCV, Hugging Face Ecosystem.
 - **Evaluation & Benchmarking:** p50/p90/p95/p99 latency percentiles, F1/Precision/Recall, VRAM tracking, Pytest CI/CD.
 - **Cloud & DevOps:** AWS (EC2, S3, ECS, Fargate, App Runner), GCP, Docker, GitHub Actions CI/CD.
 - **Programming:** Python (Expert), Flutter/Dart, React Native, Node.js, SQL/PostgreSQL.
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Education & Certifications

Education

- **B.Sc. in Computer Science** | Rizvi College, Mumbai University (2017 – 2020) — CGPA: 9.33 / 10.00
- **Vocational Computer Science / HSC (12th)** | M.H. Saboo Siddik Polytechnic, Mumbai (2015 – 2017) — 75.80%

Certifications

- **AI & Emerging Tech:** AI & Quantum Computing Mastery, Machine Learning with Python, Complete Quantum Computing Course.
- **Healthcare Compliance:** Healthcare API Compliance & HIPAA PHI Data Governance.
- **FinTech Standards:** FinTech Regulatory Standards & PCI-DSS Payment Processing.